

NEUTRINO OSCILLATION EXPERIMENT CONTROL

WHAT IS A NEUTRINO?

Neutrinos are small particles that only interact with other particles through the weak force.

There are three flavours of neutrino: electron, muon and tau.

ν_τ

ν_e

ν_μ

ν_1

WHY DO THEY OSCILLATE?

Each flavour of neutrino is a mixture of the three different neutrino mass states. As they travel, the mixture of mass states changes slightly, so when the neutrino is

ν_2

ν_3

finally detected, there is a chance that it will have changed flavour.

HOW ARE THEY MEASURED?

The Deep Underground Neutrino Experiment will send neutrinos 1300km through the earth from Fermilab to Sanford Underground Research Facility – by checking how many neutrinos of different flavours and energies arrive at Sanford, particle physicists can learn more about the difference between the mass states, the difference between neutrinos and anti-neutrinos, and how much of each mass is in each flavour mixture.

WHY DOES THIS MATTER?

Learning more about neutrino oscillations will help to answer fundamental physics questions such as:

WHY IS THE
UNIVERSE MADE OF
MATTER AND NOT
ANTIMATTER?

HOW DO BLACK
HOLES AND
SUPERNOVAE
OCCUR?

WHAT IS
BEYOND THE
STANDARD MODEL
OF PHYSICS?

HOW TO PLAY

1. CHOOSE YOUR PARAMETERS

DM32

Make mass state 3 bigger or smaller than mass state 2

TH23

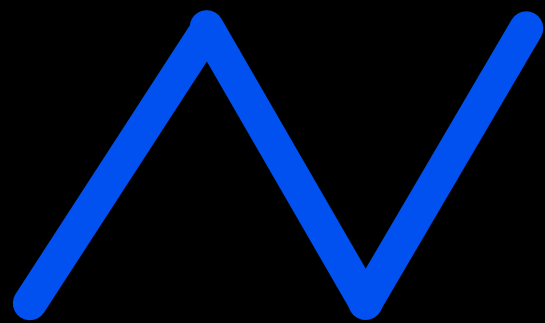
Change how much of each mass state is in the beam

DCP

Make anti-neutrinos behave differently to neutrinos

BONUS: +1X FOR EVERY PARAMETER

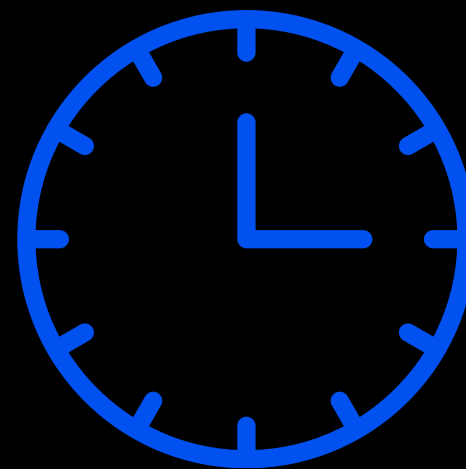
2. CHOOSE YOUR MODES



Noise

Detectors aren't perfect — sometimes they miss neutrinos or make them up.

BONUS: 1.5X SCORE

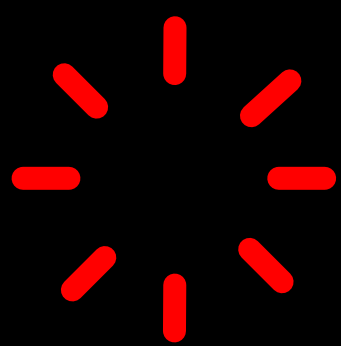


Real Time Data

Neutrino events are really rare — try to match your model to the data while the new events occur

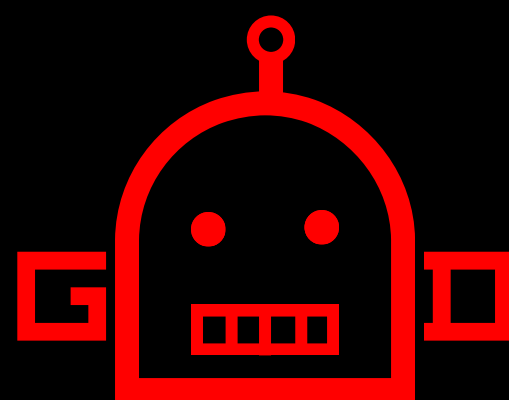
BONUS: 2X SCORE

3. CHOOSE YOUR AI OPPONENT



NONE

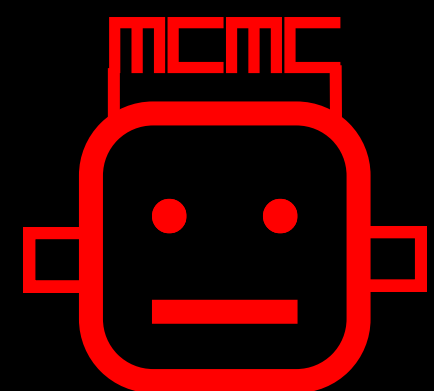
Just you vs the clock. No distractions.



GRADIENT DESCENT

Constantly jumps in the direction of a better answer, trying to find the lowest likelihood spot it can reach

BONUS: 2X SCORE

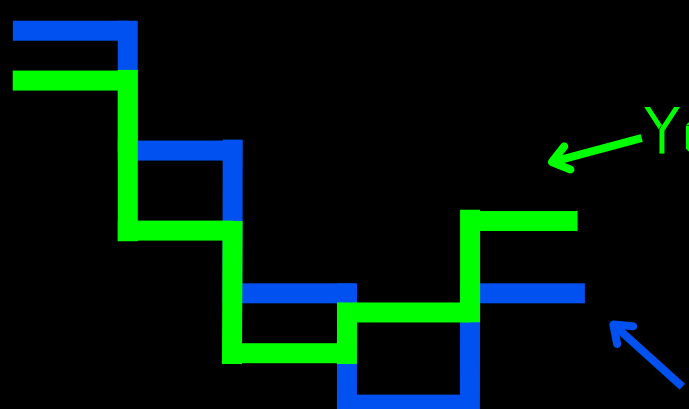


MARKOV CHAIN MONTE CARLO

Stalks around the space, picking new spots based on a combination of their likelihood and random chance.

BONUS: 2X SCORE

4. MATCH THE DATA



Change the parameters until your model matches the data from the detector — then hit the big red button to finalise your model and get your score!

To get the highest score, get as close to the data as you can, as fast as you can!

